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Case Report

Left hepatectomy for hepatic hydatid cyst with intra-biliary rupture: Better to be radical

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ABSTRACT

Echinococcal liver cysts are predominantly located in the right lobe of the liver and are mostly asymptomatic. A frank intra-biliary rupture (IBR) of hydatid cyst is uncommon, having variable clinical presentation and treatment options. We present a case of a 60-year-old male patient who presented with pain in the upper abdomen associated with vomiting but without jaundice. On investigations, he was diagnosed to have a left lobe hepatic hydatid cyst (HHC) with IBR for which left hepatectomy with bile duct exploration was performed. It highlights the benign nature of the disease for which seldom major hepatectomies have to be performed.

Introduction

Hydatid cyst is predominantly found in the liver and detected incidentally on sonography for other causes. Medical management alone seldom suffices and surgery is the gold standard for symptomatic hydatid liver disease.[1] The benign nature of the disease usually calls for a conservative surgical approach rather than radical resection. However, complicated cysts such as those presenting with intra-biliary rupture (IBR), secondary infection, and suppuration may pose a clinical challenge.[2]

Radical resections, though uncommon for this benign entity, offer high success rates with regards to the extirpation of the parasite with anatomical resections offering less morbidity in terms of bile leaks and other adverse effects such as recurrence, hemorrhage, and cholangitis and can be resorted to, in selected scenarios.[3] Here, we discuss a rare case of left lobe hepatic hydatid cyst with intra-biliary rupture managed with a radical approach.

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Case report

A 60-years-old male patient presented with dull aching pain in the right upper abdomen for the last three months. It was associated with non-bilious vomiting for five days. There was no history of jaundice, fever, melena, hematemesis, or loss of weight. Abdominal examination revealed a lump in the epigastric region with its upper margin indistinct from the liver.

Ultrasonography of the abdomen showed a large heterogeneous echogenic lesion containing membranes in the left lobe of the liver suggestive of a hydatid cyst. Hydatid cyst membranes were extending into the common bile duct (CBD) via the left hepatic duct (LHD). Magnetic resonance imaging (MRI) abdomen confirmed the hydatid cyst of size 97 × 90 mm in the left lobe of the liver involving segments II, III, and IV with multiple T2 hypointense membranes. Cysto-biliary communication (CBC) was observed with LHD and membranes were extending into CBD with upstream biliary radical dilatation (Fig. 1a–d). The left lobe was relatively atrophied. His liver function tests were grossly normal.

Endoscopic retrograde cholangiography (ERC) revealed a submucosal bulge in the antrum along the lesser curvature and multiple hydatid cyst membranes were seen in the CBD beside the left lobe hydatid cyst but no obvious CBC. ERC clearance was endeavored, however, because of incomplete clearance, a plastic biliary stent was placed in the left ductal system and surgical intervention was planned. He was started on oral albendazole two weeks prior to surgery.

Intra-operatively, a 10 × 10 cm hydatid cyst was seen involving segments II, III, and IVb of the Liver with atrophic segment IVa was documented using an intra-operative ultrasound. Along with it, CBC with the LHD was confirmed. Left hepatectomy with clearance of the membranes and biliary stent from CBD was performed via the LHD and ascertained using choledochoscopy before closing the LHD primarily (Fig. 2a–d).

The post-operative period was uneventful and he was discharged by post-operative day 4 on albendazole which was continued for one month. The patient continues to do well at two years follow-up with no recurrence as evidenced by clinical symptomatology and six-monthly ultrasound scans as well as six monthly IgM and IgG serology by ELISA.

Discussion

Hepatic hydatid cysts (HHCs) are mostly asymptomatic (60%) unless they achieve a large size (> 10 cm) or complications arise, which occurs in almost one-third of cases.[1,4] HHC is most commonly solitary (70%) and is seen in the right lobe of the liver in 80% of cases with isolated involvement of the left lobe being seen only in one-fifth of cases.[2,4,5] The incidence of IBR of HHC has been reported in 3–17% and it occurs in the right hepatic duct (RHD) in 55–60%, into LHD in 25–30% of cases and remaining, rupture into the confluence or gall bladder.[5,6]

The causative factors of IBR include trauma, secondary infection, or increasing pressure due to an increase in the size of the cyst.[7] Rupture more likely occurs in deep-seated cysts when an intra-cystic pressure exceeds 80 cm of H_2O.[5] There are three types of hepatic cyst ruptures: contained, communicating, and

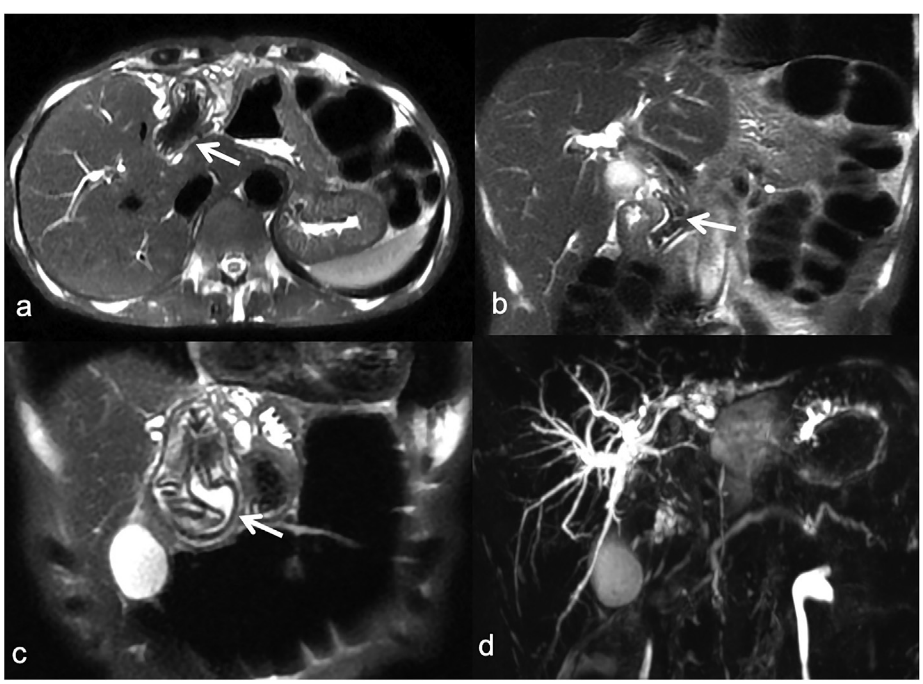

Fig. 1 – Magnetic resonance imaging, T2 weighted images: a. Cysto-biliary communication with the left hepatic duct in axial section (arrow). Left lobe atrophy is also evident; b. T2 hypointense membranes and contents in the common bile duct shown in the coronal plane (arrow); c. coronal image showing hydatid cyst (arrow) that contains T2 hyperintense fluid with hypointense membranes; d. 3D-MRCP image shows bilateral intrahepatic biliary radical (IHBR) dilatation with crowding of IHBR in the left lobe.

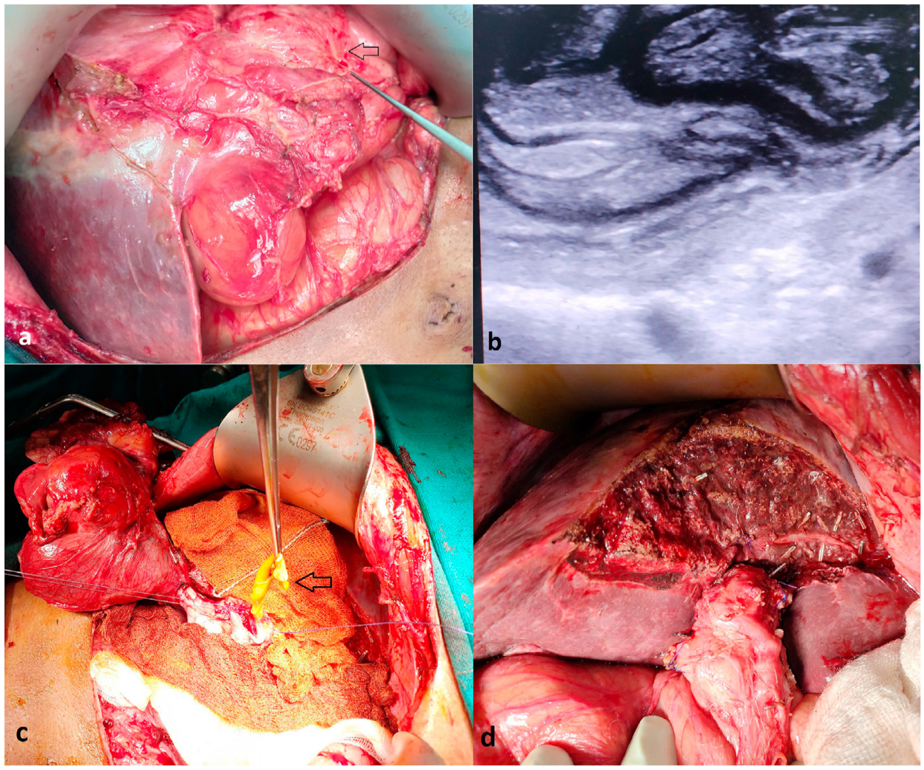

Fig. 2 – Intra-operative images: a. Showing hydatid cyst involving segment II and IVb of the liver (shown by arrow); b. intra-operative ultrasound demonstrating hydatid cyst in the left lobe of liver; c.taking out hydatid membranes (denoted by arrow) from the common bile duct via the opening in the left hepatic duct; d. showing the remnant liver post left hepatectomy.

direct. *Contained rupture* involves disruption of endocyst with an intact pericyst, demonstrated by detached membranes floating in the hydatid fluid on ultrasound; *communicating rupture* involves perforation of pericyst causing egression of daughter cysts into bile ducts and the ensuing bile duct obstruction; *Direct rupture* involves a breach of both endocyst and pericyst causing spillage of daughter cysts into pleural or peritoneal cavities]. However, only the communicating type may be considered as true IBR, as in our case.[4,8]

The diagnosis is based on epidemiology, presence of pathognomonic features on imaging, and serology. IBR may present as biliary colic, obstructive jaundice, cholangitis, or rarely liver abscess.[7] Ultrasound or computed tomography findings can be used to make the diagnosis with sensitivities of 85% and 100% for IBR.[9] Complications of HHC especially IBR can be better documented with MRI with Magnetic resonance cholangiopancreatography (MRCP) and managed by ERC.[4] Delayed diagnosis and treatment of IBR are associated with increased morbidity (19–43%) and mortality (1.8–4.5%).[10]

The management of symptomatic patients is primarily surgical with peri-operative albendazole therapy given to reduce the viability of cysts at the time of surgery and reduce the incidence of cyst recurrence.[11] The choice of approach depends on the patient and disease factors (performance status, number, and location of cysts, type of cysts, concomitant complications) and also surgeon factors (experience, skill, and availability of intensive care).[3] The radical procedures include excision of the intact cyst alone or along with the part of the liver around it which is known as cystopericystectomy. In context to HHC with CBC, operative approach encompasses clearing mother's cyst of membranes, clearance of CBD along with meticulous identification and suturing of CBC if identified.[11] On the other hand, lesser invasive procedures intent at deroofing the cyst and managing the remnant cavity with omentoplasty, marsupialization, or capitonnage.[12] Apart from this, percutaneous approaches in the form of puncture, aspiration, injection, re-aspiration (PAIR) have shown success in a carefully selected cohort and are contraindicated in patients with IBR.

Our case was unique in the view that hydatid cyst membranes were present in CBD yet there was no evidence of jaundice or cholangitis, which has been reported in 57–100% or 70–90% cases respectively with hydatid cyst rupture into the biliary tree.[3] The HHC underwent a communicating IBR into the LHD which is seen in nearly one-fourth of cases and further required a left-hepatectomy which constitutes less than 7% of all liver resections done for HHC.[13] Another learning lesson is that even a well-performed ERC can miss the CBC in IBR and hence in all such cases it should be actively sought intra-operatively and closed to prevent post-operative bile leaks. Furthermore, we emphasize the fact that HHC is a benign disease for which radical surgeries in the form of major liver resections may be required in selective cases like ours. Hence, the surgeon, based on sound judgment and clinical skills should be ready for anatomical hepatectomies. It has been shown beyond doubt that radical surgeries in the form of hepatectomies are better tolerated by these patients with less morbidity.[1,2] Moreover, less radical procedures are associated with a high risk of bile leak, biliary fistula, cholangitis, and recurrence of infection sometimes requiring reintervention but these may be the only option in patients unfit for major surgeries or in the elder patients.[14]

In conclusion, HHC along with IBR represents a typical subset of patients in whom timely intervention may prevent life-threatening complications. Formal liver resection is warranted in such selected scenarios and may be better tolerated with respect to recurrence, complications, and ensure a better quality of life.

Consent

Informed patient consent was obtained for publication of the case details.

Disclosure of competing interest

The authors have none to declare.

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